$3d^6$

August 18, 2026

1 $\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{x^2-y^2}^\alpha 3d_{z^2}^0\right\rangle$

Here are the CSFs:

$$|1,2,2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|2,2,2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|3,2,2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|4,2,2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|5,2,2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$\begin{aligned}
|6,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{2}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{2}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle \\
|7,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{2}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle \\
|8,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{2}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle \\
|9,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{2}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle
\end{aligned}$$

$$
\begin{aligned}
|10,2,1\rangle =& \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|11,2,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|12,2,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
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\begin{aligned}
|13,2,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|14,2,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|15,2,0\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|16,2,-1\rangle =& \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|17,2,-1\rangle =& \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|18,2,-1\rangle =& \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|19,2,-1\rangle =& \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|20,2,-1\rangle =& \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$|21,2,-2\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|22,2,-2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|23,2,-2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|24,2,-2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|25,2,-2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|26,1,1\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|27,1,1\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|28,1,1\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|29,1,1\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|30,1,1\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$
\begin{aligned}
|31,1,0\rangle =& + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$|32,1,0\rangle = +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|33,1,0\rangle = +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|34,1,0\rangle = +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|35,1,0\rangle = +\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$
$$- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$
$$|36,1,-1\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|37,1,-1\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|38,1,-1\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|39,1,-1\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|40,1,-1\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$|41,1,1\rangle = +\sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|42,1,1\rangle = +\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|43,1,1\rangle = +\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|44,1,1\rangle = +\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|45,1,1\rangle = +\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$
$$- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|46,1,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|47,1,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|48,1,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|49,1,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|50,1,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$+ \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$
\begin{aligned}
|51,1,-1\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
|52,1,-1\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
|53,1,-1\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
|54,1,-1\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle\\
|55,1,-1\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle\\
&-\sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle\\
|56,1,1\rangle =&\frac{\sqrt{3}}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|57,1,1\rangle =&\frac{\sqrt{3}}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|58,1,1\rangle =&\frac{\sqrt{3}}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|59,1,1\rangle =&\frac{\sqrt{3}}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|60,1,1\rangle =&\frac{\sqrt{3}}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\
&-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\
|61,1,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|62,1,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|63,1,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|64,1,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|65,1,0\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle\\
&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\
|66,1,-1\rangle =&\frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle\\
&+\frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{\sqrt{3}}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$|67,1,-1\rangle = \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|68,1,-1\rangle = \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|69,1,-1\rangle = \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|70,1,-1\rangle = \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{\sqrt{3}}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|71,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|72,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|73,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|74,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|75,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|76,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|77,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|78,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|79,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|80,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|81,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|82,1,1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|83,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|84,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|85,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|86,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|87,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|88,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|89,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|90,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|91,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|92,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|93,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|94,1,1\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|95,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|96,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|97,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|98,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|99,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|100,1,1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|101,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|102,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|103,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|104,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|105,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|106,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|107,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|108,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|109,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|110,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|111,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|112,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|113,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|114,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|115,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|116,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|117,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|118,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|119,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|120,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|121,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|122,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|123,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|124,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|125,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|126,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|127,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|128,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|129,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|130,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|131,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|132,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|133,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|134,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|135,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|136,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|137,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|138,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|139,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|140,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|141,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|142,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$|143,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$
$$|144,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|145,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$
$$|146,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$
$$|147,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|148,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$|149,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$
$$|150,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$
$$|151,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$
$$|152,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|153,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$|154,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$
$$|155,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|156,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|157,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$|158,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$|159,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$|160,1,-1\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$
\begin{aligned}
|161,0,0\rangle =& + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|162,0,0\rangle =& + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|163,0,0\rangle =& + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|164,0,0\rangle =& + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|165,0,0\rangle =& + \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
|166,0,0\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|167,0,0\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|168,0,0\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$|169,0,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|170,0,0\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$
$$- \frac{1}{\sqrt{12}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$
$$|171,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$
$$|172,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|173,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|174,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$
$$|175,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|176,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$
$$|177,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$
$$|178,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|179,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$
$$|180,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|181,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$
$$|182,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|183,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|184,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|185,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|186,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|187,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|188,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|189,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|190,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|191,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|192,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|193,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|194,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|195,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|196,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|197,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|198,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|199,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|200,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|201,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|202,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|203,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|204,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|205,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|206,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|207,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|208,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|209,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|210,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

The Ground State CSFs are 72 102 132

$\langle 72,1,1\,|L_x|\,26,1,1\rangle = \sqrt{\frac{3}{2}}I$
$\langle 72,1,1\,|L_x|\,41,1,1\rangle = \frac{3}{\sqrt{2}}I$
$\langle 72,1,1\,|L_x|\,77,1,1\rangle = -I$
$\langle 72,1,1\,|L_x|\,84,1,1\rangle = -I$

Δg_{xx}

$+\frac{3}{2}\zeta\Delta_{72-26}^{-1}$
$+\frac{9}{2}\zeta\Delta_{72-41}^{-1}$
$+\zeta\Delta_{72-77}^{-1}$
$+\zeta\Delta_{72-84}^{-1}$

$\langle 72,1,1\,|L_y|\,27,1,1\rangle = -\sqrt{\frac{3}{2}}I$
$\langle 72,1,1\,|L_y|\,42,1,1\rangle = -\frac{3}{\sqrt{2}}I$
$\langle 72,1,1\,|L_y|\,75,1,1\rangle = I$
$\langle 72,1,1\,|L_y|\,86,1,1\rangle = -I$

Δg_{yy}

$+\frac{3}{2}\zeta\Delta^{-1}_{72-27}$
$+\frac{9}{2}\zeta\Delta^{-1}_{72-42}$
$+\zeta\Delta^{-1}_{72-75}$
$+\zeta\Delta^{-1}_{72-86}$

For a multiplicity of 3 we have the following:

$$D = \langle 1,1|1,1\rangle - \langle 1,0|1,0\rangle$$

$$E = \langle 1,-1|1,1\rangle$$

$$\begin{aligned}\left\langle 1,1\left|\hat{H}_{eff}\right|1,1\right\rangle = -\,\zeta^2\,(\\ &+\frac{3}{4}\Delta_1^{-1}+\frac{3}{4}\Delta_2^{-1}+\frac{1}{8}\Delta_{11}^{-1}+\frac{1}{8}\Delta_{12}^{-1}+\frac{3}{16}\Delta_{31}^{-1}+\frac{3}{16}\Delta_{32}^{-1} \\ &+\frac{1}{16}\Delta_{46}^{-1}+\frac{1}{16}\Delta_{47}^{-1}+\frac{1}{8}\Delta_{61}^{-1}+\frac{1}{8}\Delta_{62}^{-1}+\frac{1}{8}\Delta_{105}^{-1}+\frac{1}{8}\Delta_{107}^{-1} \\ &+\frac{1}{8}\Delta_{114}^{-1}+\frac{1}{8}\Delta_{116}^{-1}+\frac{3}{16}\Delta_{161}^{-1}+\frac{3}{16}\Delta_{162}^{-1}+\frac{1}{16}\Delta_{166}^{-1}+\frac{1}{16}\Delta_{167}^{-1} \\ &+\frac{1}{8}\Delta_{175}^{-1}+\frac{1}{8}\Delta_{177}^{-1}+\frac{1}{8}\Delta_{184}^{-1}+\frac{1}{8}\Delta_{186}^{-1}\,)\end{aligned}$$

$$\begin{aligned}\left\langle 1,0\left|\hat{H}_{eff}\right|1,0\right\rangle = -\,\zeta^2\,(\\ &+\frac{3}{8}\Delta_6^{-1}+\frac{3}{8}\Delta_7^{-1}+\frac{3}{8}\Delta_{16}^{-1}+\frac{3}{8}\Delta_{17}^{-1}+\frac{3}{16}\Delta_{26}^{-1}+\frac{3}{16}\Delta_{27}^{-1} \\ &+\frac{3}{16}\Delta_{36}^{-1}+\frac{3}{16}\Delta_{37}^{-1}+\frac{1}{16}\Delta_{41}^{-1}+\frac{1}{16}\Delta_{42}^{-1}+\frac{1}{16}\Delta_{51}^{-1}+\frac{1}{16}\Delta_{52}^{-1} \\ &+\frac{1}{8}\Delta_{56}^{-1}+\frac{1}{8}\Delta_{57}^{-1}+\frac{1}{8}\Delta_{66}^{-1}+\frac{1}{8}\Delta_{67}^{-1}+\frac{1}{8}\Delta_{75}^{-1}+\frac{1}{8}\Delta_{77}^{-1} \\ &+\frac{1}{8}\Delta_{84}^{-1}+\frac{1}{8}\Delta_{86}^{-1}+\frac{1}{8}\Delta_{135}^{-1}+\frac{1}{8}\Delta_{137}^{-1}+\frac{1}{8}\Delta_{144}^{-1}+\frac{1}{8}\Delta_{146}^{-1} \\ &+2\Delta_{201}^{-1}+2\Delta_{203}^{-1}\,)\end{aligned}$$

$$
\begin{aligned}
\left\langle 1,-1 \left| \hat{H}_{eff} \right| 1,1 \right\rangle = &- \zeta^2 \Big(\\
&- \frac{1}{8}\Delta_{11}^{-1} + \frac{1}{8}\Delta_{12}^{-1} + \frac{3}{16}\Delta_{31}^{-1} - \frac{3}{16}\Delta_{32}^{-1} + \frac{1}{16}\Delta_{46}^{-1} \\
&- \frac{1}{16}\Delta_{47}^{-1} + \frac{1}{8}\Delta_{61}^{-1} - \frac{1}{8}\Delta_{62}^{-1} - \frac{1}{8}\Delta_{105}^{-1} + \frac{1}{8}\Delta_{107}^{-1} \\
&+ \frac{1}{8}\Delta_{114}^{-1} - \frac{1}{8}\Delta_{116}^{-1} - \frac{3}{16}\Delta_{161}^{-1} + \frac{3}{16}\Delta_{162}^{-1} - \frac{1}{16}\Delta_{166}^{-1} \\
&+ \frac{1}{16}\Delta_{167}^{-1} + \frac{1}{8}\Delta_{175}^{-1} - \frac{1}{8}\Delta_{177}^{-1} - \frac{1}{8}\Delta_{184}^{-1} + \frac{1}{8}\Delta_{186}^{-1} \\
&\Big)
\end{aligned}
$$

$$
\begin{aligned}
D = &\zeta^2 \Big(\\
&- \frac{3}{4}\Delta_{1}^{-1} - \frac{3}{4}\Delta_{2}^{-1} + \frac{3}{8}\Delta_{6}^{-1} + \frac{3}{8}\Delta_{7}^{-1} - \frac{1}{8}\Delta_{11}^{-1} - \frac{1}{8}\Delta_{12}^{-1} \\
&+ \frac{3}{8}\Delta_{16}^{-1} + \frac{3}{8}\Delta_{17}^{-1} + \frac{3}{16}\Delta_{26}^{-1} + \frac{3}{16}\Delta_{27}^{-1} - \frac{3}{16}\Delta_{31}^{-1} - \frac{3}{16}\Delta_{32}^{-1} \\
&+ \frac{3}{16}\Delta_{36}^{-1} + \frac{3}{16}\Delta_{37}^{-1} + \frac{1}{16}\Delta_{41}^{-1} + \frac{1}{16}\Delta_{42}^{-1} - \frac{1}{16}\Delta_{46}^{-1} - \frac{1}{16}\Delta_{47}^{-1} \\
&+ \frac{1}{16}\Delta_{51}^{-1} + \frac{1}{16}\Delta_{52}^{-1} + \frac{1}{8}\Delta_{56}^{-1} + \frac{1}{8}\Delta_{57}^{-1} - \frac{1}{8}\Delta_{61}^{-1} - \frac{1}{8}\Delta_{62}^{-1} \\
&+ \frac{1}{8}\Delta_{66}^{-1} + \frac{1}{8}\Delta_{67}^{-1} + \frac{1}{8}\Delta_{75}^{-1} + \frac{1}{8}\Delta_{77}^{-1} + \frac{1}{8}\Delta_{84}^{-1} + \frac{1}{8}\Delta_{86}^{-1} \\
&- \frac{1}{8}\Delta_{105}^{-1} - \frac{1}{8}\Delta_{107}^{-1} - \frac{1}{8}\Delta_{114}^{-1} - \frac{1}{8}\Delta_{116}^{-1} + \frac{1}{8}\Delta_{135}^{-1} + \frac{1}{8}\Delta_{137}^{-1} \\
&+ \frac{1}{8}\Delta_{144}^{-1} + \frac{1}{8}\Delta_{146}^{-1} - \frac{3}{16}\Delta_{161}^{-1} - \frac{3}{16}\Delta_{162}^{-1} - \frac{1}{16}\Delta_{166}^{-1} - \frac{1}{16}\Delta_{167}^{-1} \\
&- \frac{1}{8}\Delta_{175}^{-1} - \frac{1}{8}\Delta_{177}^{-1} - \frac{1}{8}\Delta_{184}^{-1} - \frac{1}{8}\Delta_{186}^{-1} + 2\Delta_{201}^{-1} + 2\Delta_{203}^{-1} \\
&\Big)
\end{aligned}
$$

$$
\begin{aligned}
E =\zeta^2 \Big(& \\
& + \frac{1}{8}\Delta_{11}^{-1} - \frac{1}{8}\Delta_{12}^{-1} - \frac{3}{16}\Delta_{31}^{-1} + \frac{3}{16}\Delta_{32}^{-1} - \frac{1}{16}\Delta_{46}^{-1} \\
& + \frac{1}{16}\Delta_{47}^{-1} - \frac{1}{8}\Delta_{61}^{-1} + \frac{1}{8}\Delta_{62}^{-1} + \frac{1}{8}\Delta_{105}^{-1} - \frac{1}{8}\Delta_{107}^{-1} \\
& - \frac{1}{8}\Delta_{114}^{-1} + \frac{1}{8}\Delta_{116}^{-1} + \frac{3}{16}\Delta_{161}^{-1} - \frac{3}{16}\Delta_{162}^{-1} + \frac{1}{16}\Delta_{166}^{-1} \\
& - \frac{1}{16}\Delta_{167}^{-1} - \frac{1}{8}\Delta_{175}^{-1} + \frac{1}{8}\Delta_{177}^{-1} + \frac{1}{8}\Delta_{184}^{-1} - \frac{1}{8}\Delta_{186}^{-1} \\
& \Big)
\end{aligned}
$$